

- AI-driven threats: automated phishing, polymorphic malware, adversarial ML, misuse of generative AI

Unit II - Securing AI Systems

- Model, data, and pipeline security controls
- Protecting training data, model weights, and inference endpoints
- Secure AI deployment in cloud, on-prem, and hybrid infrastructures
- Threats: model theft, inversion, poisoning, evasion, membership inference
- Privacy-preserving AI methods: differential privacy, federated learning, synthetic data
- Secure design principles for AI system architecture

Unit III - AI-Assisted Security

- AI-enabled anomaly and threat detection
- SOC workflow automation: event triage, alert correlation, response orchestration
- Behavior analytics and UEBA
- AI-driven threat modeling and continuous monitoring

Unit IV - Governance, Risk, Compliance, and Privacy

- Regulatory and governance frameworks: GDPR, EU AI Act, NIST AI RMF, ISO/IEC AI standards
- AI governance and accountability structures
- Risk management across the AI lifecycle
- Responsible AI: ethics, fairness, transparency, explainability
- Documentation and auditability: model cards, data sheets

Assessments:

- Midterm Exam - 30% (October 1, 2026)
- Research Project and Paper - 30% (Presentation by Nov 19, 2026)
- Final Exam - 30% (Set by the university)
- Quiz and Participation - 10%

Textbook

- Lecture handouts will be provided based on different sources and textbooks.

Recommended Resources:

- Ian Goodfellow, Deep learning. (2016).
- NIST AI 100-2: Adversarial Machine Learning- A Taxonomy and Terminology of Attacks and Mitigations
- Miller, David J., Zhen Xiang, and George Kesidis. Adversarial Learning and Secure AI. Cambridge University Press, 2023.
- NIST AI Risk Management Framework
- NIST SP 1270: Adversarial Machine Learning
- MITRE ATLAS
- IEEE Ethics Guidelines
- Research papers from IEEE S&P, USENIX, NeurIPS, ICML, ACL, ICLR, etc.

Instructional Method:

Lectures, discussions, readings, and case-based analysis. This is a lecture-based course but a few lab activities will be included.